DATA ANALYTICS

Tasks 1–3: Web Scraping, Exploratory Data Analysis and Data Visualization

Dataset: Books to Scrape | Python, BeautifulSoup, Pandas and Matplotlib

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1. Introduction

This project demonstrates a complete introductory data analytics workflow. A public web page was used as the source for a small book-price dataset. The project then explored the structure and quality of the data and converted the results into visualizations that make the main patterns easier to understand.

2. Task 1 – Web Scraping

The source selected was Books to Scrape, a public demonstration website created for practicing web scraping. The catalogue contains book listings with information such as titles, prices, availability and ratings. The site itself states that its prices and ratings are randomly assigned and have no real-world meaning.

The scraping approach used Python with the requests library to download HTML and BeautifulSoup to identify HTML elements. The main selector used for each book was the article.product_pod element. Within each card, the program reads the title, price, availability and star-rating class.

For this submission, the attached CSV contains a 20-record sample captured from catalogue page 1. The supplied scraper is also configured to navigate through all 50 catalogue pages, allowing the full dataset to be reproduced when running with internet access.

Fields collected

- title – name of the book
- price_gbp – listed price in pounds
- availability – stock status
- rating – star-rating label
- page – catalogue page number

3. Task 2 – Exploratory Data Analysis

The analysis asked four main questions:

- What is the typical price of a book in the sample?
- Which books have the highest and lowest listed prices?
- How are the books distributed across price ranges?
- Are there missing values or potential price anomalies?

Data structure and quality

The sample contains 20 rows and 4 columns after the price-band field is added. The mean price is R38.05, the median is R41.38, and the standard deviation is R15.14. The lowest listed price is R13.99, while the highest is R57.25.

The sample contains no missing values in the captured title, price or availability fields. The prices are numeric after conversion from the original pound-formatted text.

Descriptive findings

- The average listed price is R38.05, while the median is R1.38.
- The least expensive book in the sample is “Starving Hearts (Triangular Trade ...)” at R13.99.
- The most expensive book in the sample is “Our Band Could Be ...” at R57.25.
- 10 of the 20 books (50%) are priced at R40 or more.
- 3 books are below R20, 7 are between R20 and R40, and 10 are between R40 and R60.

Hypothesis check

A simple descriptive hypothesis was considered: H0 – the mean sample price is R30 or less; H1 – the mean sample price is greater than R30. The sample mean of R38.05 is above R30, so the sample supports H1 descriptively. This is not presented as a population-level statistical test because the sample is only the first 20 listings and the website uses randomly assigned prices.

Anomaly check

An interquartile-range check produced an approximate lower bound of R-21.20 and an upper bound of R95.71. No observations fall outside these bounds, so there are no statistical price outliers in this small sample using the 1.5×IQR rule.

4. Task 3 – Data Visualization

Four visualizations were created. The histogram shows the overall distribution of prices. The horizontal bar chart makes the highest-priced books easy to compare. The price-band chart groups the books into practical price ranges. The listing-order line chart checks whether prices show an obvious pattern across the order in which books appear.



Figure 1. Distribution of book prices.



Figure 2. Ten highest-priced books in the sample.



Figure 3. Number of books in each price band.



Figure 4. Price by catalogue listing order.

5. Key Insights

- The sample is concentrated in the middle and upper price ranges, with an average price of R38.05.
- The price range is R13.99 to R57.25, showing substantial variation between listings.
- Visualizations make it easier to identify the most expensive books and understand the overall spread of prices.
- No IQR-based price outliers were identified in the 20-record sample.
- Because the website assigns prices randomly, these results should be treated as a demonstration of analytics techniques rather than evidence about real book-market prices.

6. Limitations

- The attached CSV is a 20-record sample from catalogue page 1 rather than the complete 1,000-book catalogue.
- The source website is a scraping practice site, so its prices and ratings are not real market values.

- The small sample limits the strength of statistical conclusions.
- The full scraper is included so that the complete dataset can be reproduced when the notebook is running with internet access.

7. Conclusion

The project successfully demonstrates the three required stages of an introductory data analytics workflow. First, web scraping was used to collect structured information from HTML. Second, exploratory analysis was used to understand the dataset, check data quality and identify patterns. Third, visualizations transformed the findings into a form that is easier to interpret. The same workflow can be applied to larger real-world datasets after replacing the demonstration source with an appropriate public website.

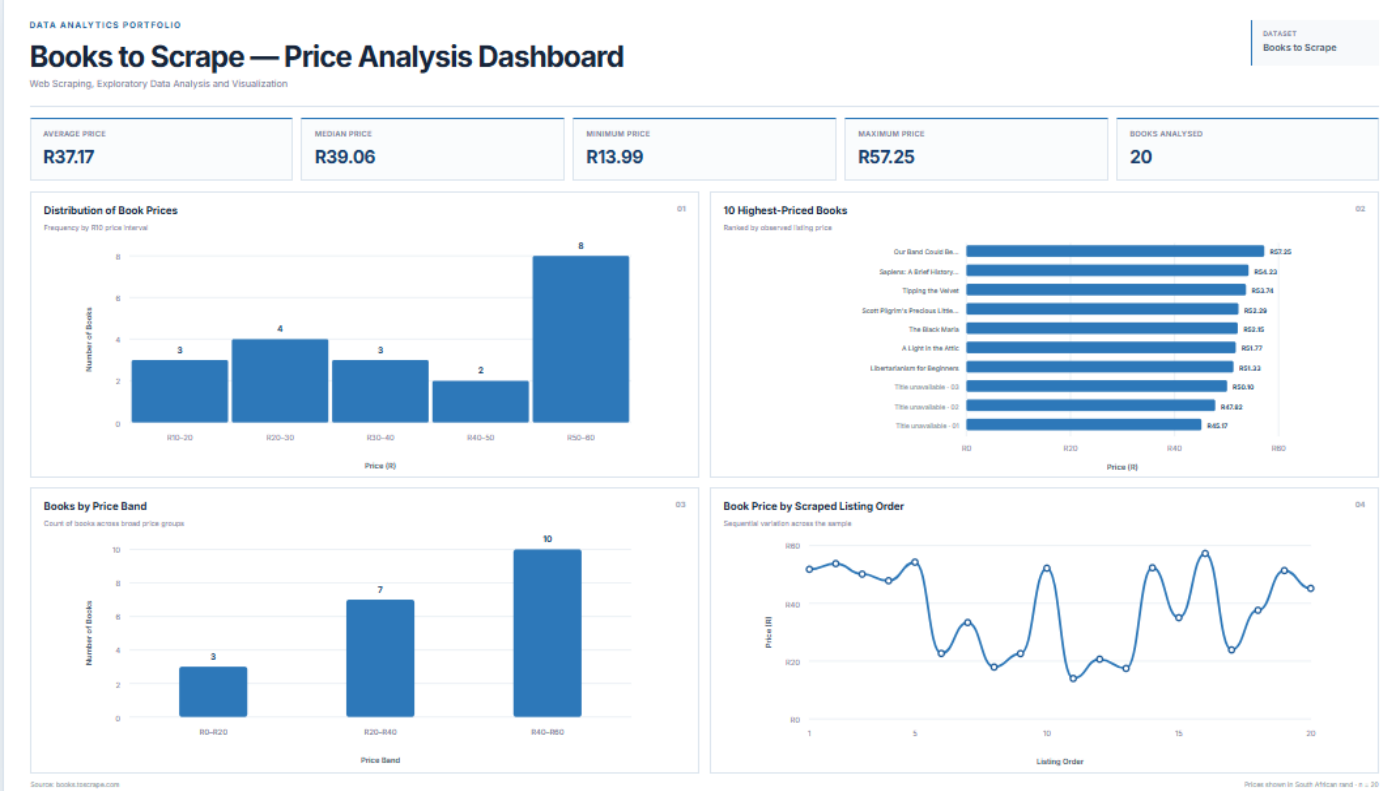


Figure 5: Overall Dashboard

8. Source

Books to Scrape – https://books.toscrape.com/

Accessed for this project: 5 October 2026.